WORKSHEET · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 3: Project: Search Showdown

Module 1 | 50 minutes | Project: comparing algorithms | Grades 6–8 · ages 12–14

Name: _____ Class: _____ Date: _____

Plan your project and practise the key ideas.

1. Key words

Match each word to what it means. Write the letter.

a. fair test ___ A. A test to compare performance.
b. efficiency ___ B. How much work an algorithm needs.
c. benchmark ___ C. Changing only the thing you're comparing.

2. Which search fits?

Write each item in the right group: Linear search / Binary search.

a. A shuffled deck of cards → _____
b. A dictionary → _____
c. Today's 5 unsorted messages → _____
d. A sorted list of 1 million user IDs → _____
e. A list you can't sort → _____

3. My plan

Plan your project: Run linear and binary search on the same sorted list of 100 numbers, count the steps each takes for several targets, and report which is more efficient — with each search in its own procedure. Sketch it and list what each part will do.

4. Checklist

Tick each one when your project does it.

a. ☐ Sorted list of 100
b. ☐ Linear search with steps
c. ☐ Binary search with steps
d. ☐ Procedures
e. ☐ Both reported
f. ☐ Results table

5. Two stars and a wish

Ask a friend to test your project. Write two things they liked and one idea to improve it.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I compared two algorithms fairly.
☐ I organised them into procedures.
☐ I drew a conclusion from evidence.